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Chapter 2

Nurturing the seeds of transfer: a domain-specific perspective[☆]

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Abstract

Transfer, the process of using knowledge or skills acquired in one context in a new or varied context, has long been the topic of spirited debate in the research community. In this chapter, fundamental dimensions that appear to underlie effective transfer are described. These fundamentals — subject-matter knowledge, general cognitive and metacognitive strategies, and motivation, particularly interest — are referred to as the seeds of transfer. As a means of presenting these fundamentals, five generalizations about transfer, drawn from the extensive literatures in transfer, domain learning, and analogical reasoning, are summarized. In addition, potential instructional barriers to transfer are highlighted, along with critical steps deemed necessary to enhance transfer within academic settings. © 1999 Elsevier Science Ltd. All rights reserved.

The argument against transfer becomes more believable when one realizes that universities are full of people who are attempting to make one significant transfer. They are called professors. (Detterman, 1993, p. 2)

Accordingly, it is worth noting that there are vast areas in which transfer is not a problem at all and where it is so straightforward that no one bothers to do research on it. (Bereiter, 1995, p. 26)

Transfer, the process of using knowledge acquired in one situation in some new or novel situation, has long been a quandary of Gordian proportions for applied psychologists. As the opening quotes suggest, transfer has also been the topic of

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spirited debate within the research community (Cox, 1997; Goska & Ackerman, 1996). On the one hand, there are those, like Detterman, who view the evidence on transfer and shake their heads in utter despair, claiming that cases of significant transfer are “rarer than volcanic eruptions and large earthquakes” and as difficult to predict (1993, p. 2). On the other hand, there are those, like Bereiter (1995), who argue for a less grandiose notion of transfer, and who, thus, see clear and frequent evidence of transfer in the learning of fundamental processes (e.g., reading) and in the application of conceptual learning across contexts.

In reconciling the contrasts in these two perspectives, Gentner and Markman (1997) offer their own comparison, abandoning Detterman’s colorful “volcanoes and earthquakes” metaphor in favor of one referencing comets and planets. According to Gentner and Markman, the normatively significant insights that Detterman (1993) describes are like comets — riveting, brilliant, and rare flashes of perspicacity that streak through one’s mind. In contrast, there are those more prosaic, highly predictable, and less dramatic occasions of transfer that oftentimes go unnoticed. These, according to Gentner and Markman, are like planets that move about the heavens with predictable and quiet regularity. Despite their disparate appearances and varied behaviors, as Gentner and Markman are quick to point out, both comets and planets are regulated by similar fundamental principles and both are central astronomic phenomena.

Transfer, like comets and planets, takes multiple forms with differing probabilities of occurrence. Yet, whether one is on the *low road* (i.e., literal similarity, reflexive action, and well-practiced tasks) or the *high road* (i.e., abstraction of principles, effortful processing, increased complexity, and novelty) to transfer (Salomon & Perkins, 1989), the fundamental processes remain relatively consistent. As noted, transfer involves the process of applying one’s prior knowledge or experience to some new situation. Further, the tremendous range of opinions within the psychological and educational communities as to the meaning, nature, and likelihood of transfer notwithstanding, there are at least five defensible generalizations about transfer.

1. Generalizations about transfer

Transfer is an essential aspect of competent performance in any complex domain. There is ample evidence in the literature that highly competent learners, experts (and possibly some professors) regularly perform tasks for which they have never been directly trained (Chi, Feltovich & Glaser, 1981; Corkill & Fager, 1995; Lerda, Garzunel & Therme, 1996). For example, a skilled professor might use her understanding of social hierarchies as a mechanism for understanding the societal workings and hierarchy of various ant colonies. Similarly, a learner who is competent in physical science will apply his knowledge of pulleys, levers, and fulcrums to ease the burden of moving heavy boxes out of his garage and into a truck. However, the same cannot be said for learners who are less competent or only acclimated in their fields.

As James (1890) so duly noted, such individuals do not possess the excess consciousness required to grapple with the transfer of knowledge. Essentially, the less

competent science learner must expend his cognitive energies on grasping the theory behind centrifugal force rather than using that information to understand why he does not fall out of a loopy-loop roller coaster. Further, our research would suggest that these acclimated learners also lack the strategic ability and interest requisite for carrying out transfer in a complex domain (e.g., Alexander, Jetton & Kulikowich, 1996; Alexander, Murphy, Woods, Duhon & Parker, 1997). Thus, it would appear that transfer is a critical dimension of able performance. Moreover, even if Detterman's admonition — "if you want people to learn something, teach it to them" (1993, p. 21) — were heeded, it is impossible to convey even the most basic of academic knowledge and skills in all their potential iterations. No teacher, regardless of how dedicated, effective, or creative, can deal with the time demands and complexity (to say nothing of the tedium) that such instruction would necessitate.

Transfer of schooled knowledge and procedures occurs far less often than educators and educational researchers would hope. The literature is replete with testimonials to the relative ineffectiveness of "training for transfer" studies (e.g., Brown, Collins & Duguid, 1989; Salomon & Perkins, 1989). That is to say, students who are taught to do *X* under conditions *Y* frequently fail to do *X* under conditions *Z*. Suppose, for example, a fifth grader is taught how to represent a one-step comparison word problem graphically as a way to decide the appropriate algorithm to employ. When encountering word problems that involve multiple steps or alternative functions, however, the child does not attempt to transform the linguistic problem into graphic form.

When studies have been successful at producing transfer, it is far more likely to have been *near* transfer than *far* transfer. For instance, Singley and Anderson (1989) reviewed the theoretical and empirical transfer literature from Thorndike to the present and found very little documentation of far or general transfer. They concluded: "Besides the spate of negative evidence, there has been no positive evidence of general transfer besides a few highly questionable studies" (p. 25). In general, training studies tend to demonstrate transfer effects when: (a) researchers or practitioners target near transfer tasks rather than far transfer tasks; (b) the structural similarities are still rather apparent or literal; or, (c) helpful hints and cues prompt students to use trained knowledge or skills (Detterman, 1993).

In addition, Perkins (1992) and others (e.g., Campione, Shapiro & Brown, 1995; Holyoak & Thagard, 1997) have insisted that successful transfer is dependent on a sense of purpose in the content to be learned, as well as the situations in which the knowledge can and should be transferred. That is to say, students should be engaged in meaningful, problem-based activities for which the knowledge and skills acquired are means for accomplishment rather than ends in themselves. Further, students should be shown how the knowledge and skills they attain in one problem-based situation have utility in other relevant contexts. Such conditions are often difficult to achieve in classrooms as they are presently configured. Therefore, even if an optimistic outlook on the occurrence of transfer is assumed, the situation is still far from optimal. There is much more that can and should be done to create a disposition toward transfer among learners (Bereiter, 1995).

Transfer is more apt to occur when the learning environment is intentionally orchestrated to encourage cross-situation and cross-domain transfer and rewards such efforts.

There are literally volumes dedicated to teaching for transfer. Among the characteristics that are regularly discussed under this rubric are adequate time devoted to training, explicitness of instructional content, attention to basic problem-solving procedures or strategies, multiple and varied opportunities for transfer, and mindful reflection and abstraction of the materials to be taught (e.g., McKeough, Lupart & Marini, 1995; Pressley, 1995). Still, as noted, these reasonable characteristics have not often resulted in instructional outcomes that are readily transferable to more than the most similar of tasks or that endure long after training terminates (Detterman, 1993). These disappointing findings underscore that teaching for transfer is very demanding and requires a long-term instructional commitment, adequate resources and learning tasks, as well as knowledgeable and dedicated teachers (Cox, 1997; Marini & Genereux, 1995).

Beyond these reasonable components, transfer seems to require a level of enculturation into the domains from which concepts and problems arise (Greeno, Moore & Smith, 1993). Yet, the nature of schooling is customarily such that students do not have the opportunity to develop a realistic or deep sense of the academic domains that are the mainstay of formal education. Accordingly, because of their piecemeal or patchwork understanding of such fields as mathematics, history, or science, students continue to treat concepts or procedures in a fragmented or unduly constrained manner (Byrnes, 1996). In essence, students see each task as a separate exercise, an isolated performance event where maximum points are often earned merely for duplicating what was taught in just the manner instructed. Moreover, since the failure to transfer is rarely associated with lower grades in classrooms, students feel little compunction when transfer is not evidenced.

Transfer and analogical reasoning are related processes. In its most general sense, analogical reasoning is the ability to establish relationships between two seemingly dissimilar entities (Corkill & Fager, 1995; Schunn & Dunbar, 1996). Detterman's (1993) likening of transfer to volcanic eruptions is an illustration of analogical thinking in that it entails comparing ideas from one domain (psychology) to concepts in another, quite varied, domain (geophysics). The value of such a process is that it allows individuals to reconceptualize and reposition more abstract or novel ideas (e.g., transfer) in terms of concepts that are perhaps more concrete and more familiar (e.g., volcanoes). For this process to work effectively (if at all), such analogical reasoning requires that individuals possess conceptual understanding of both the source and target domains (e.g., psychology and geophysics). Further, these domains must be linked through certain relational processes (e.g., inferring or mapping; Pirolli & Recker, 1994; Sternberg & Rifkin, 1979).

The investigations of various researchers give credence to this view that the processes underlying conceptual transfer correspond directly to those required to reason analogically (Gentner & Markman, 1997; Reeves & Weisberg, 1994). So much so, in fact, that many researchers investigate transfer as though it were a special case of analogical reasoning (e.g., Kolodner, 1997; Reeves & Weisberg, 1993). Certainly, it can be argued that if learners fail to see the similarities between particular tasks or contexts, then it is improbable that they will engage in the mindful transfer of conceptual or procedural knowledge across tasks or contexts.

The seeds of transfer are multifaceted, encompassing aspects of the learner, the content, and the context. In considering the challenge of teaching for transfer, Marini and Genereux (1995) asked the thought-provoking question of whether educators should focus their efforts on tasks, learners, or contexts in their instruction? They summarily state that “taking all three into account when designing instruction is most advisable” (p. 5). Even thoughtfully designed instructional programs cannot hope to succeed often or well unless they attend simultaneously to the *learner*, the *content*, and the *context*. This claim is based on our research in academic development and analogical reasoning, as well as on the informative studies of others (e.g., Ackerman & Heggestad, 1997; Goska & Ackerman, 1996; Lupart, 1995). Furthermore, the interplay among particular learner attributes establishes the fertile ground for transfer and for the analogical thought that underlies that transfer. In the discussion that follows, certain learner, content, and context variables that build the foundation for transfer are discussed briefly.

2. Identifying the seeds of transfer

For the past five years, the authors have focused their energies on understanding the transformations that occur in students as they engage in domain-specific learning. One outcome of this research is the model of domain learning or MDL (see Alexander, 1997). In the MDL, the systematic changes that individuals undergo as a result of formal learning in a domain (e.g., physics, history, or reading) are theorized to occur in three stages: acclimation, competency, and proficiency/expertise. The distinctions among acclimated, competent, and proficient learners are predicated on changes in their subject-matter knowledge, their situational and individual interests, as well as their general strategic (cognitive and metacognitive) processing. These same dimensions of knowledge, interest, and strategic processing guide our examination of transfer, as well.

2.1. Stages of domain learning

Learners in acclimation have limited and fragmented knowledge of the domain (Chi et al., 1981), including declarative (what), procedural (how), and conditional (when and where) understanding (Paris, Lipson & Wixson, 1983; Ryle, 1949). Further, their interests are relegated more by the aspects of the immediate situation or context (i.e., situational interest; Schiefele, 1992) than to any enduring, personal identification with or valuing of the domain (i.e., individual interest; Hidi, 1990). In addition, their unfamiliarity with the domain means that they encounter many problems that are novel or complex for them. Thus, they must rely heavily on general cognitive and metacognitive strategies to aid them in their learning and performance. However, their lack of relevant subject-matter knowledge also means that their strategic processing is often ineffective or inefficient (Alexander & Judy, 1988).

Competent learners, by comparison, not only have acquired quantitatively more domain knowledge than acclimated learners, but that knowledge is also more

principled in form (Gelman & Greeno, 1989). That is to say, the subject-matter knowledge of competent learners is coherently organized around key domain concepts and principles. Moreover, competent learners manifest an increasing interest or investment in the domain and its related topics, and are simultaneously less dependent on situationally interesting dimensions in the immediate context to motivate their learning and development. Finally, through the process of enculturation, problems typical of the academic domains become more routine for competent learners, allowing for a more efficient interplay between domain-specific and general strategic processing during this stage.

Lastly, for the few who achieve proficiency, there is a tremendous breadth and depth of subject-matter knowledge. Another hallmark of proficiency is knowledge creation. In essence, proficient individuals contribute new knowledge to their field, transforming that domain in some fashion. Also, experts have a powerful and abiding interest in the domain and identify strongly with the community of practice. In addition, because these experts are pushing the boundaries of their domain and are engaged in problem formulation, they show an increased employment of general cognitive and metacognitive strategies over competent learners. However, unlike acclimated learners who often rely on lower-order or surface-level strategies (e.g., rereading or rehearsal), proficient learners tend to use more higher-order strategies (e.g., synthesis or summarization) that allow for a depth of processing (Alexander, Graham & Harris, 1998a). Among the various strategies studied is the fundamental ability to establish relationships across seemingly unrelated phenomena or analogical reasoning (Sternberg & Rifkin, 1979).

2.2. Domain-specific analogical reasoning

Our study of analogical reasoning has focused primarily on the ability to solve classic domain-specific analogy problems (Alexander, Jetton & Kulikowich, 1996; Alexander & Murphy, 1998b). The solution of such domain-specific analogy problems requires not only an understanding of underlying cognitive processes, but also a body of domain-specific knowledge, as well as a willingness to engage in effortful processing. These problems are of the form A:B::C:D, and consist of terms drawn from a specific academic domain. Two examples of these problems from the fields of human biology and educational psychology, respectively, are *red blood cells:erythrocytes::leucocytes:white blood cells*, and *Pavlov:classical conditioning::Skinner:operant conditioning*. Given that analogical reasoning and transfer are such similar processes, this research can broaden our understanding of transfer and the conditions that promote such thinking.

2.3. Profiles of academic achievement

From research on domain-specific learning, diverse student profiles associated with academic achievement and emerging learner competence have been identified. These profiles bear a strong resemblance to others described in the literature on student achievement (e.g., Recker & Pirolli, 1995; Renkl, 1997). In one of these studies

particularly relevant to this discussion the external criterion variable used to validate student profiles was performance on a domain-specific analogy task (Alexander & Murphy, 1998b). Specifically, after students participated in a semester of instruction in educational psychology, they formed four distinct clusters that differed significantly with regard to domain-specific analogical reasoning ability. The four clusters were labeled the learning oriented (LO), strong knowledge (SK), effortful processors (EP), and nonstrategic reader (NR).

Students in the LO cluster ($n = 94$) displayed high knowledge and high interest in the domain and employed deeper-level strategies (e.g., summarizing or determining main idea) during text processing. LO students also received the highest scores on the domain-specific analogy task at posttest. By comparison, students in the SK cluster ($n = 39$) were significantly less interested and less strategically engaged in the domain than those in the LO cluster. They were also less effective than LO students at learning from a demanding domain-specific reading task. Still, their scores on the domain-specific analogy task, while lower than LO students at posttest, were not statistically different. It is important to note, however, that SK students began the semester with significantly more domain-related knowledge than students in the other three clusters, and apparently relied on this pre-existing knowledge to carry them through the term.

The EP ($n = 76$) and the NR ($n = 120$) clusters encompassed our less academically successful students. Those in the EP group were distinct in their low domain knowledge scores, their moderate domain interest, and their moderate level of performance on the text-processing and domain-specific analogy tasks. This cluster, however, evidenced the most strategic effort during text processing, although much of that effort involved rather surface-level strategies (e.g., rereading parts of the text or changing rate of reading). In effect, this learner profile was unique in the amount of effort exerted for a limited academic payoff.

Finally, those in the NR cluster, the largest grouping of students at semester's end, garnered only a moderate level of domain knowledge from instruction, had only a moderate level of interest in the content, engaged in less strategic processing, and, not surprisingly, recalled the least from the text-processing task. Performance on the domain-specific analogy measure was similarly rated as low to moderate.

Based on this and other related research, there are three primary ingredients that contribute to students' academic achievement that configure in different ways over the course of domain learning. They are domain knowledge, motivation (e.g., interest), and general strategic ability. Moreover, students who show the greatest growth toward domain competence and perform best on domain-specific analogy tasks exhibit high levels of all three of these ingredients. Should any one of these components be weakly developed, students' academic progress over time could well be jeopardized. The one exception was the Strong Knowledge cluster. These students, like Renkl's (1997) anticipative reasoners, seem to rely heavily on an existing base of knowledge to carry them, despite limited interest and investment in the domain. It remains doubtful, however, whether this initial base of domain knowledge would prove sufficient to propel students into competence once they move beyond such an initial foray and into more advanced domain exploration.

This configuration of knowledge, strategic processing, and interest came into play again in an investigation of error patterns on domain-specific analogy measures (Alexander, Murphy & Kulikowich, 1998). In this study, the emphasis went beyond global judgments of right and wrong to the information conveyed in students' incorrect responses to analogy problems. To guide this analysis, a hierarchical error model devised by Alexander (1990) for classic domain-specific analogy problems of the form A:B:C: ? was used. In this model, the lowest forms of errors (Category 1, No Response; Category 2, Repetition) were indicative of limited domain and strategy knowledge. That is, students who committed these types of errors scored low on measures of domain knowledge and strategic processing. Commission of such errors, especially No Response errors, also seemed related to limited student engagement or effort. The highest forms of errors (Category 5, Domain Specific; Category 6, Target Variant), by comparison, were associated with relatively high levels of domain and strategy knowledge.

Another interesting result was that the types of errors to which individual students gravitated were very consistent (Experiment 1, Alexander et al., 1998a). In fact, for nearly two-thirds of the sixth graders studied, 50% or more of their errors fell in a single-error category. Thus, the influence of domain knowledge, strategic processing, and motivation carried well beyond the raw score students received on a domain-specific analogy task to the types of mistakes they committed. Further, these errors were patterned and were differentially indicative of respondents' domain knowledge, strategic processing ability, and seemingly their motivations for task performance. Based on these studies, domain knowledge, strategic processing, and motivation (e.g., interest) underlie competent performance in academic domains, including the ability to transfer knowledge and skills within and across that domain (i.e., analogical reasoning).

3. Recognizing potential instructional barriers to transfer

Elsewhere the authors have written of the negative conditions within schools that seem to impede students' development of competence in academic domains, and, with it, their ability to engage in transfer (Alexander & Murphy, 1998a,b; Alexander, Murphy & Woods, 1996). These conditions function as impediments because they interfere with students' ability to construct a rich base of domain knowledge, and to devise or employ general strategies for deeper processing. These impediments also decreased students' motivations for engagement. The specific conditions are: mentioning versus teaching, instructional enabling, and decontextualized instruction.

3.1. Mentioning versus teaching

Durkin (1978–1979) admonished the educational community for their failure to provide meaningful reading instruction to elementary students. Specifically, in their efforts to “cover the content”, teachers invested far too little time exploring important knowledge and strategies richly and deeply. Instead, she noted, “mentioning” stood in

lieu of teaching for far too much of the instructional day. Sadly, there is little reason to assume that the pattern Durkin observed 20 years ago has improved. If anything, the amount of content to be covered has only multiplied in the past two decades.

The problem of superficial treatment of subject matter is not new, of course. In *How We Think*, Dewey (1910/1991) decried that: “In school, amassing information always tends to escape from the ideal of wisdom or good judgment. ‘Covering the ground’ is the primary necessity; the nurture of the mind a bad second” (p. 52). Questions have also been raised about the concepts that teachers emphasize (or even mention) in classrooms and whether central domain-specific concepts are given any more weight instructionally than more tangential notions (e.g., Griffin, Case & Capodilupo, 1995). That is, it is not simply of concern that teachers may *mention* rather than *teach*, but that what gets mentioned may be of questionable importance to building a body of principled conceptual knowledge within any given domain.

In several investigations, for example, Alexander and her colleagues (Alexander, Jetton, Kulikowich & Woehler, 1994; Jetton & Alexander, 1997) determined that teachers often spent instructional time on content that was high in interest value but low in subject-matter importance. Garner, Gillingham and White, (1989) suitably labeled these high interest/low importance ideas as “seductive details.” Jetton and Alexander (1997), who studied high-school science teachers, concluded that teachers with strong domain knowledge and, at least, moderate pedagogical knowledge were more likely to focus on important domain concepts and to convey those concepts effectively to students than teachers weak in either domain or pedagogical knowledge. As long as teachers persist in treating content superficially or devote important instructional time to seductive details rather than to key domain concepts, then students are less likely to form the principled body of subject-matter knowledge needed to progress into competence or to demonstrate transfer.

3.2. *Instructional enabling*

In a review of research on learning environments, Garner (1990) surmised that students more often engage in strategic processing when they are placed in classrooms where strategic effort is required and rewarded. Regrettably, Alexander and Knight (1993) argued that many teachers have instituted a practice of *instructional enabling*. Instructional enabling, according to Alexander and Knight, entails the tacit agreement forged between teachers and students whereby students are not academically challenged but are given high marks if they perform well on relatively undemanding tasks, display compliance, and behave as good citizens in the classroom. In short, it would be Alexander and Knight’s contention, and that of others (e.g., Garner, 1990), that classrooms are not particularly academically challenging environments where strategic efforts are routinely warranted or rewarded.

Even in those instances where teachers have incorporated strategy instruction in their classrooms, presumed strategies are taught more like skills (Alexander, Graham & Harris, 1998a). That is, students are presented with a strategy, be it domain-specific or general, as if it were a *fait accompli*, a precise set of steps to be executed in just the manner in which they were taught and then ritualistically practiced on a set of

well-specified tasks until automaticity is achieved. Such “[s]heer imitation, dictation of steps to be taken, mechanical drill”, as Dewey (1910/1991) warns, “may give results most quickly and yet strengthen traits likely to be fatal to reflective power” (p. 51). For strategy instruction to contribute to transfer, students must be: (a) shown the value of such effortful processing; (b) encouraged to modify and personalize the general heuristic; and, (c) provided with opportunities and incentives for such time-consuming and mentally taxing reflection. Consequently, as long as the incentives for students to personalize strategies or to go beyond the instructional parameters remain limited and, as long as the time and rewards for strategic effort are in short supply, transfer either within or across domains is unlikely to be stimulated.

3.3. *Decontextualized and disconnected instruction*

The phrase “piecemeal” has been applied to classroom learning (Garner and Alexander, 1989), and appropriately so. The associations that are made in what students learn from one day to the next — to say nothing of one domain to the next or one year to the next — are minimal (e.g., Perkins, 1992; Salomon & Perkins, 1989). One reason that the first author felt strongly about the need for a model of domain learning is that both teachers and their students seemed to move through the educational experience without any well articulated map for what could or should transpire. Understanding how experts in various fields think and behave may offer a vague picture of what a learner might look like at the end of a long and fruitful educational journey (e.g., Chi, Glaser & Farr, 1988; Ericsson & Smith, 1991). However, it provides little guidance as to how the “end” is to be reached and how learners systematically change over the course of such an extended trek.

It is not merely that students do not see much of a connection between the world outside of the classroom and what they are learning, although, sadly, this is often the case (Dewey, 1910/1991; Kolodner, 1997). The potential relationships between and among the concepts and procedures students acquire within and across domains are rarely articulated and rarely inferred (Dienes & Berry, 1997). For this reason, students can complete their compulsory educational period with diplomas, and possibly good grades, in hand and never appreciate how their learning serves any goal other than the completion of the next assignment or the passing of the next exam. Certainly, students have little opportunity to become enculturated into these academic domains — to see biology with the eyes of a biologist or to experience history as an historian (Ball, 1993; VanSledright, 1996). Instead, they think as students who acquire bodies of inert knowledge that retain their viability only for as long as they are needed to satisfy some immediate and short-lived tasks (Whitehead, 1957).

Why we, as educational researchers, are so surprised that transfer does not occur more often or that students have not formed a disposition toward transfer is a conundrum. Students are merely exemplifying the very behaviors that are commonly modeled and routinely reinforced. Periodic training, even when effectively devised and delivered, is not likely to transform the habits of the mind (James, 1890) that have been formed and reinforced over hundreds, and potentially thousands, of instructional episodes.

4. Nurturing transfer

In light of the aforementioned barriers to transfer, several alterations to learning environments should be made that will enhance the likelihood of transfer. First and foremost, a domain-specific perspective should be taken to the problem of transfer. This is important because the seeds of transfer are more likely to take root when those in the classroom community see themselves as progressing toward competence in specific fields of study. As with apprenticeship models of learning (e.g., Brown & Campione, 1990; Rogoff, 1990), the goal is more long-term than the next assignment, the next test, or the next report card. Teachers and students share in the objective of “rooted relevance” (Alexander et al., 1996) whereby schooled information is not only meaningfully connected to learners and their worlds outside of the classroom, but also to the fundamental concepts and procedures that define particular domains or disciplines (Ball, 1993, Bereiter, 1995; diSessa, 1989).

Also, understanding the roles that analogical reasoning, in general, and transfer, in particular, play in this journey toward emerging competence helps to give students’ purpose to such effortful processing (Holyoak & Thagard, 1997). That is to say, the following changes, nested within a domain-specific framework, may stimulate knowledge and strategy development in students, and may contribute to students’ personal investment in the target domain. Moreover, these suggestions may give rise to a classroom context where learning is challenging, knowledge and knowing are valued, and the forging of associations across tasks and contexts is practiced and prized.

4.1. *Stage a three-pronged attack*

Our first recommendation is a general admonition to those who care about transfer. As is evident to many who labor in the field of applied psychology, the problem of transfer is multifaceted. It is not a question of whether efforts should be focused on the learner, the task, *or* the context (Marini & Genereux, 1995), or whether the solution to the transfer problem rests in knowledge, strategy, *or* motivational training. All are parts of the problem and, consequently, parts of the solution. To assume that any training program that attends only to one of these elements or that only temporarily offsets the prevailing educational forces that deter transfer will somehow transform those who rarely or non-spontaneously evidence cross-task and cross-domain reasoning seems delusional.

Dispositions toward transfer (Bereiter, 1995), at a minimum, require a rich and cohesive body of domain knowledge, a well-honed strategic repertoire, as well as a personal investment in or identification with an academic domain. These seeds, unlike Jack’s magic beans, cannot spring forth overnight into some monumental structure capable of transporting students into the realm of transfer. Only a long-term commitment to nurturing all three of these seeds has any real hope of producing students with a propensity toward transfer.

This three-pronged approach must also be tailored to fit the inter-individual and intra-individual differences that exist at varying stages in their academic development

(Ackerman, Kyllenon & Roberts, 1999; Alexander, 1997). For instance, students who are acclimated with regard to a particular domain are understandably more reliant on conditions within the immediate context to stimulate their involvement or to promote their learning. These acclimated learners are also more in need of a foundation of relevant subject matter knowledge upon which to base subsequent development (Alexander, 1997).

As cluster-analytic studies have shown, there are also those who project varying configurations of knowledge, interest, and strategic processing that may necessitate differing instructional responses (e.g., Recker & Pirolli, 1995; Renkl, 1997). For example, those like the Effortful Processors profiled in Alexander and Murphy (1998b) may require particular attention to their strategic processing by learning how to apply strategies effectively and efficiently. Nonstrategic Readers, in contrast, may need assistance in recognizing the merits of purposeful strategic engagement to their continued academic development.

4.2. Strive for principled understanding

Those who study transfer frequently proclaim the power of knowledge (e.g., Byrnes, 1996; Corkill & Fager, 1995; Schunn & Dunbar, 1996). Yet, educators must not mistake a simple accumulation of knowledge with the call for principled knowledge made here. Piecemeal or inert knowledge, common among many students, contributes little to transfer. What is required is a cohesive body of knowledge that can serve as an anchor for cross-task or cross-domain transfer (Murphy & Woods, 1996). When students possess principled knowledge in a domain, the central concepts within this domain can function as the source or grounding for analogical reasoning. For this principled knowledge to take form, however, two conditions must exist.

First, educators guiding students' domain learning must know their fields well enough to make thoughtful decisions about what concepts and procedures should become the centerpieces of classroom instruction (Alexander et al., 1996). This is as true for those teaching young children as those working with the upper grades. The seeds for transfer must be planted early in the educational process. Important knowledge and skills must be placed center stage during instruction, with more tangential content appropriately given only minor or cameo roles. When educators recognize the relative value of domain knowledge, they may be more likely to convey the importance of this knowledge to their students. Further, when teachers have a principled understanding of a domain, they have a focal point in their instruction. Therefore, they may be more inclined to make linkages among concepts and to relate subsequent content to those key principles, thereby modeling the processes and illustrating the significance of transfer.

Second, if they strive for principled understanding, educators must set aside any inclination to mention rather than teach. Perhaps, as diSessa (1989) recommends, teachers should adopt the agenda of teaching more about less. With sufficient instructional time to play with key concepts, teachers may be better able to address the individual differences found within the classroom. When educators are free to teach more about less they can also revisit central notions under varying conditions

and in diverse fashions — factors that have been shown to promote transfer (Dienes & Berry, 1997). For instance, the application of this knowledge must be practiced in diverse contexts to promote low-road transfer, and students must be encouraged and rewarded for mindfully reflecting on those concepts as a mechanism for promoting high-road transfer (Perkins, 1992; Salomon & Perkins, 1989).

4.3. *Model and reward analogical thinking*

Finally, educators should not expect students to display a disposition toward transfer when they, themselves, cannot be described as models of transfer. As Alexander and her colleagues have determined (Alexander et al., 1994; Jetton & Alexander, 1997), students are relatively effective at determining what is valued within the classroom context and are surprisingly strategic at altering their behavior to conform to this “instructional importance”. As Dewey (1910/1991) stated: “His [the student] chief concern is to accommodate himself to what the teacher expects of him, rather than to devote himself energetically to the problems of subject-matter” (pp. 49–50).

As such, teachers must model analogical thought and transfer as a regular course of their teaching and expect it from their students. We do not agree with Detterman (1993) that educators must assume that transfer is as rare as volcanic eruptions and should operate as though it is virtually unachievable as an instructional goal. Rather, teachers should forge an environment that is conducive to analogical reasoning and transfer and should likewise reinforce and reward evidence of transfer, whether low-road or high-road, whenever it is exhibited. In addition, teachers should address the purpose of transfer, as well as the value of the knowledge and tasks being learned.

Detterman’s (1993) ultimate solution to the problem of transfer was the admonition: “If you want people to learn something, teach it to them”. In contrast, our admonition to educators would be: “If you want students to engage in analogical thinking and to manifest transfer, then become the very models of those processes”. Teachers’ occasioning of transfer and their acknowledgment of cross-task and cross-domain reasoning within the classroom community should be as commonplace and as predictable as the movements of the planets (Gentner & Markman, 1997). When that becomes the case in our educational systems, then such debates and extended discussions of transfer, as represented by this issue, may become as rare as major eruptions.

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